

Lymphatic & Immune Systems Laboratory

The lymphatic system is a diverse group of organs, tissues and vessels that serves many functions in the body. It's an important part of the immune system that helps defend us from many harmful agents in the internal and external environments. It works with the cardiovascular system to maintain fluid homeostasis and with the gastrointestinal system to absorb our dietary fats. We can list 3 primary functions of the lymphatic system and organs:

1. Transports excess interstitial fluid back to the heart
2. Absorbs dietary fats
3. Activates the immune system

In completing the activities for this section, you'll study just how the immune system does all these amazing things for the human body.

MATERIALS

- human torso or chart with lymphatic organs or use Real Anatomy
- compound microscope, lens paper
- prepared slides of the lymph nodes, lymph vessel/valve, and spleen
- Diagrams provided by instructor
- Construction paper, scissors, glue and/or tape, colored pencils

Important Definitions:

Lymph: the fluid found within lymphatic vessels, formed from blood plasma

Lymphocytes: A lymphocyte is a type of white blood cell that is part of the immune system. There are two main types of lymphocytes: B cells and T cells. **The B cells** produce antibodies that are used to attack invading bacteria, viruses, and toxins. **The T cells** destroy the body's own cells that have themselves been taken over by viruses or become cancerous.

Primary lymphatic locations: contain stem cells that produce lymphocytes and are the site where these lymphocytes become immunocompetent (recognize and mount immune system responses)

Secondary lymphatic organs & tissues: sites for defense against invading agents and cancer cells and include the lymph nodes, spleen, and lymphatic nodules.

Metastasis: the spread of a disease-producing agency (such as cancer cells) from the initial or primary site of disease to another part of the body

Antigen: A substance that has immunogenicity (the ability to provoke an immune response) and reactivity (the ability to react with antibodies or cells that result from an immune response).

Antibody: a protein produced by plasma cells in response to a specific antigen. The antibody combines with that antigen to neutralize, inhibit or destroy it. Antibodies are sometimes called immunoglobulins. Since antibodies circulate in the serum, they are often called anti-(X) serum e.g. anti Rh serum would be an antibody that binds to Rh factor.

Agglutinogens: antigens found on the surface of erythrocytes (red blood cells) that cause agglutination (a reaction in which particles (such as red blood cells or bacteria) suspended in a liquid collect into clumps and which occurs especially as a serologic response to a specific antibody)

Phagocyte: a type of cell within the body capable of engulfing and absorbing bacteria and other small cells and particles.

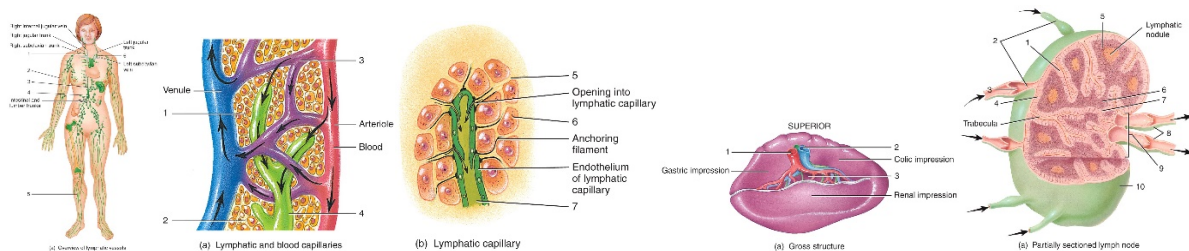
LAB ACTIVITY 1 – Take Inventory of the Lymphatic System

Either working solo or with one laboratory partner, identify (find) all of the following structures of the lymphatic system on the models in lab: (Note you'll need to use multiple models to find them all)

Lymph vessel:
Lymph nodes:
Spleen:
Tonsils:
Thymus:

LAB ACTIVITY 2 – Completing Diagrams

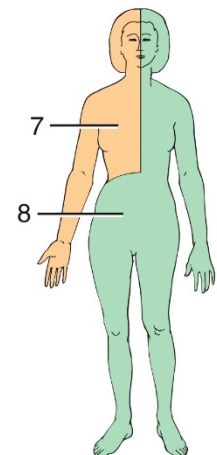
Your laboratory instructor will have placed 3-5 diagrams at this station. Please complete all diagrams, using your textbook/other resources provided to help you complete them



LAB ACTIVITY 3 – LYMPH FLOW AND THE ASYMMETRY OF THE LYMPHATIC DUCTS

Lymphatic capillaries flow into lymphatic vessels → Lymphatic vessels follow the pathway of veins in the body → Take lymph towards neck region → Lymph then flows through series/chain of lymph nodes → Lymph vessels then merge to form lymph trunks

- A. What part of the body is drained of lymph by the right lymphatic duct?
 - a. Where does this duct drain?
- B. What part of the body is drained of lymph by the thoracic duct?
 - a. Where does this duct drain?



(b) Lymph drainage

LAB ACTIVITY 4 – TRACE THE FLOW OF LYMPH

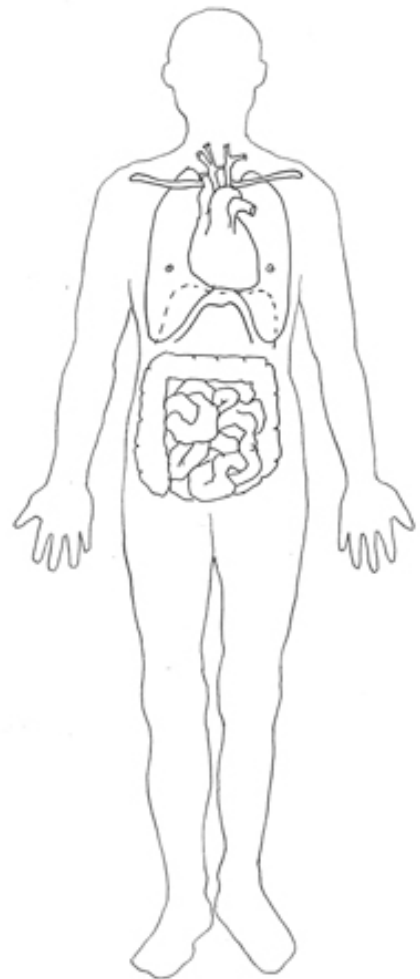
In this exercise, you will trace the pathway of lymph flow from starting location to the point at which the lymph is delivered into the cardiovascular system. Write out your pathway as you follow the lymph flow through major lymph-collecting vessels, trunks and ducts.

1. Write out the sequence of the flow
2. Then use differently colored pencils/markers to draw the pathway on the figure below.
3. Trace the flow from the following locations:

Start: RIGHT FOOT

Start: RIGHT ARM

Start: INTESTINES (WITH FAT)



LAB ACTIVITY 5 – Lymphatic Organ Histology

In this exercise you will examine the microscopic anatomy of the spleen, lymph node and thymus.

1. Review the slide of the spleen.

The largest lymphatic organ, the **spleen**, filters blood of foreign organisms and particles, eliminates aged erythrocytes, and is a blood reservoir for platelets. The spleen is the main organ that filters blood, like the lymph nodes being the main organs that filter the lymph.

Describe the Red Pulp and the White Pulp? What different functions do they do?

2. Review the slide of the lymph node.

Make a drawing representing what you see in the space below/your note book:

Include labels for:	Capsule	Trabeculae
	Afferent vessels	Efferent vessels
	Cortex	Medulla
	Hilum	Lymphatic nodule

3. Review the slide of the lymph vessel and valve.

Make a drawing representing what you see in the space below/your note book:

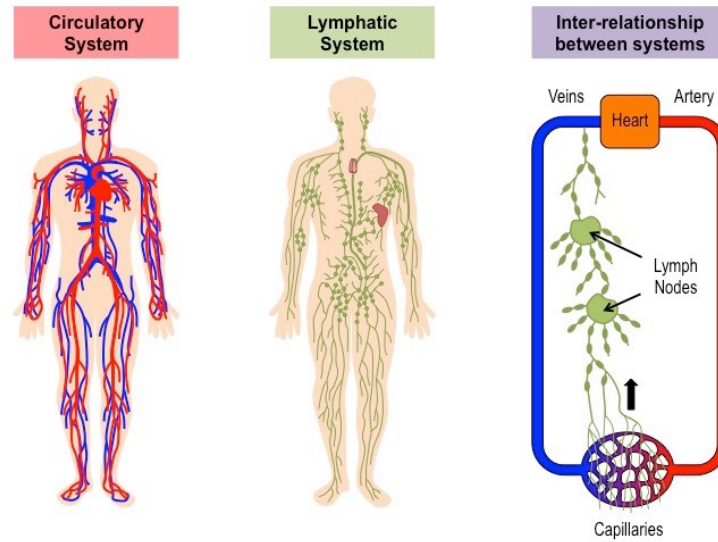
Label the valve and name the type of cell making up the wall of the vessel. Where does the fluid that this vessel would carry come from?

LAB ACTIVITY 6 – CLINICAL CONNECTIONS

1. Lymph vessels route lymph fluid through nodes throughout the body. **Lymph nodes** are small structures that work as filters for harmful substances. They contain immune cells that can help fight infection by attacking and destroying germs that are carried in through the lymph fluid. When there's a problem in your body, like an illness or an infection, your lymph nodes can become inflamed/enlarged. It's a sign that more lymphocytes are in action than usual, trying to kill off germs/foreign substances from your body. You may notice this most often in the glands in your neck (that's why your doctor feels the area under your submandibular region. The doctor is checking to see if those glands are bigger than usual or tender). **Name/discuss 3 things (diseases/infections etc) that can make your lymph nodes swell (this is called lymphadenopathy)?**
2. How does cancer spread in the lymph nodes and lymphatic vessels? Briefly explain how METASTASIS of cancer and lymph nodes are connected:

LAB ACTIVITY 7 – CARDIOVASCULAR VS. LYMPHATIC SYSTEM

Describe some of the main differences between blood circulation and lymph circulation. Think about answering – how do they each pump/get to circulate? Are they open or closed ended? What components are flowing through each?



LAB ACTIVITY 8 – CREATING IMMUNE SYSTEM MODELS

For this activity, your instructor will be giving you a pathogen! Well, not a real pathogen, but rather a paper cutout representing a pathogen. Your job will be to 1) demonstrate how this pathogen could be turned into an active ingredient for a vaccine and 2) show how cells of the immune system would respond to this vaccine and subsequent invasion by the actual pathogen. Since modeling is such a critical tool in the health sciences, you will be taking a modeling approach in this activity.

Modeling Task 1

Materials: paper pathogen from instructor

Task: Review the information below from Wikipedia regarding how “inactivated” vaccines are created.

“An inactivated vaccine (or killed vaccine) is a vaccine consisting of virus particles, bacteria, or other pathogens that have been grown in culture and then killed. In contrast, live vaccines use pathogens that are still alive (but are almost always attenuated, that is, weakened). Pathogens for inactivated vaccines are grown under controlled conditions and are killed as a means to reduce infectivity (virulence) and thus prevent infection from the vaccine.[1] The virus is killed using a method such as heat or formaldehyde. The pathogen particles are destroyed and cannot divide, but the pathogens maintain some of their integrity to be recognized by the immune system and evoke an adaptive immune response.” (https://en.wikipedia.org/wiki/Inactivated_vaccine, 2019)

What could you do to your paper model to represent what would happen to turn it into an ingredient for an inactivated vaccine? How is the pathogen different from before (we obviously would not want to put an infectious pathogen into our vaccine)? Where can you identify antigens?

Modeling Task 2

Materials: inactivated vaccine ingredient from above, colored paper, pens, pencils, scissors, tape, etc.

Begin by reviewing the following video titled “The Immune Response HD Animation.” Note the names and roles of different cells and chemicals involved in this immune response.

<https://www.youtube.com/watch?v=AucZlvEv29Y>

You might additionally want to do an image search for “adaptive immune system” or “adaptive immune response” so you have a diagram of these interactions to which you can refer.

Task: Use the materials in the room to create additional paper models so that you can demonstrate the steps and cellular interactions you would expect to happen when your vaccine ingredient is introduced into the body.

- Make sure you include any important immune cell types and any major features of those cells.
- Your paper models should be cut out so they are able to move around and interact with one another! (not just drawings on a page).

- You should be able to use your models to essentially “put on a play” showing all the interactions and activities that happen when your vaccine ingredient is introduced.

Evaluation: Once you feel you have an accurate set of models representing the adaptive immune response to a vaccine ingredient, notify your instructor. Your instructor will observe your model/play to provide feedback.

Your instructor will also bring over a new copy of the original, infectious pathogen. You should be able to show what happens with your immune cell models if this infectious pathogen now enters the body!

You might wish to take cell phone pictures of your models/model interactions or make notes regarding what you created to refer to when studying!